FACULDADE DE CIÊNCIAS DA UNIVERSIDADE DO PORTO
FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

# Inference-Time Techniques for Open-Weight Language Models in Negotiation Games

**Adriano Alexandre dos Santos Machado**

WORKING VERSION

**Master in Artificial Intelligence**

Supervisor: Prof. Gil Rocha

Second Supervisor: Prof. Henrique Lopes Cardoso

June 4, 2026

# Resumo

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**Palavras-chave**: keyword1 • keyword2 • keyword3 • keyword4

# Abstract

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# Acknowledgements

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# Declaration of honour

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Adriano Alexandre dos Santos Machado

*"We make a living by what we get, but we make a life by what we give."*

Winston Churchill

# Contents

# List of Figures

# List of Tables

# List of Acronyms

**FEUP** Faculdade de Engenharia da Universidade do Porto

# Chapter 1

# Introduction

Almost everything in life depends on negotiation. From the moment we are born, we negotiate for attention and care. Developed in part through the unconscious imitation of those around us [12], negotiation requires the ability to infer others' intentions and to act under uncertainty.

Large language models (LLMs) have demonstrated impressive capabilities across a number of tasks, from code generation to creative writing and question answering. As these models are increasingly deployed in agentic scenarios—where they must interact with other agents or humans to accomplish goals—negotiation appears as a particularly interesting benchmark.

Bianchi et al. [2] introduced NegotiationArena, an open-source framework for evaluating the negotiation abilities of LLM agents across scenarios such as the Ultimatum Game, resource trading, and price negotiation. Although highly valuable, the original evaluation was limited to closed-source models (GPT-4, GPT-3.5, Claude 2.1, and Claude 2), some of which are no longer available. This limitation raises concerns regarding reproducibility and accessibility.

Techniques such as Chain-of-Thought (CoT) allow models to think step-by-step, dividing complex problems into manageable sub-problems to reduce errors and improve interpretability. Similarly, Self-Correction mechanisms (or Self-Refine) enable models to critique and improve their own outputs through an iterative process of drafting, feedback, and refinement. While these emergent abilities have demonstrated success in domains like mathematics and coding , their specific impact on negotiation has not yet been fully studied.

Beyond individual reasoning, multi-agent collaboration offers another area for improvement. Cooperative debate, where multiple agents generate, critique, and refine each other's responses, has been shown to improve factual accuracy and reasoning quality. In real-world negotiation, decisions are rarely made by a single individual; corporate acquisitions, policy agreements, and legal settlements typically involve teams that deliberate internally before engaging with the opposing side. This observation motivates the question of whether team-based negotiation — where each side is represented by a group of LLM agents that debate before responding — can lead to stronger negotiation outcomes.

This thesis investigates whether inference-time reasoning techniques and multi-agent collaboration can improve the negotiation capabilities of large language models. The work is organized around three main contributions:

- How do open-source models compare to proprietary models in competitive negotiation scenarios?
- As inference-time reasoning techniques have shown promise individually, to what extent do they affect outcomes in multi-agent negotiation contexts?
- Can team-based negotiation, where each side is represented by a group of LLM agents that deliberate before responding, lead to stronger negotiation outcomes?

The remainder of this document is organized as follows. Chapter 2 provides a literature review covering the evolution of language models, inference-time reasoning techniques, multi-agent systems, and the different interaction scenarios relevant to this work. Chapter 3 presents the detailed work plan for the upcoming months.

# Chapter 2

# Literature Review

## 2.1 Language Models

The origins of language modeling date back to Shannon [31], which modeled natural language as a stochastic process. Building on this, Bengio et al. [1] proposed one of the first Neural Language Models (NLMs) estimating n-gram probabilities using neural networks. Thereafter, NLMs based on Long Short-Term Memory (LSTM) became the standard for many natural language tasks, including machine translation, text generation, and classification [24, 34].

The introduction of the Transformer architecture [43] addressed some limitations of LSTMs, allowing parallelization during training and capturing long-range dependencies through the self-attention mechanism. Applying this architecture, Radford and Narasimhan [29] pre-trained a language model on the next-word prediction task and showed that, when followed by task-specific fine-tuning, it achieved state-of-the-art performance on common-sense reasoning, question answering and many other tasks. GPT-2 [30] demonstrated that sufficiently large language models can perform downstream tasks such as summarization, translation, and question answering without task-specific fine-tuning.

The increase in the number of parameters, dataset size, and training compute of language models continued, supported by the power-law relationship detected in Kaplan et al. [18], which showed that an increase in these three components led to an improvement in performance. As models grew, so did the computational costs and risks associated with their deployment, which led the majority of frontier labs to keep the model weights proprietary; nevertheless, some have opted for an open-weight release (e.g., LLaMA [13], Mistral [16], DeepSeek [9]). Table 2.1 compares some of the main families of models.

## 2.2 Inference-Time Reasoning Techniques

Beyond simple improvements in performance, scaling also brought new capabilities that could not be predicted by extrapolating the performance of smaller models. Wei et al. [45] refers to them as *emergent abilities*. Among them are the ability to learn from examples (Few-Shot prompting,

Table 2.1: Comparison of different models. For Mixture of Experts (MoE) models, parameters are denoted as *Total / Active*.

| **Model** | **Parameters** | **License** | **Notes** |
|---|---|---|---|
| *OpenAI* | | | |
| GPT-1 | 117M | Proprietary | [29, 30] |
| GPT-2 | 117M–1542M (4 variants) | Open-weights | [30] |
| GPT-3 | 125M–175B (7 variants) | Proprietary | [4] |
| GPT-4, GPT-5 | Undisclosed | Proprietary | [26, 33] |
| GPT-OSS | 117B/5.1B, 21B/3.6B | Open-weights | MoE [27] |
| *Google* | | | |
| Gemini 1.0–3 | Undisclosed | Proprietary | [8, 35] |
| Gemma | 2B, 7B | Open-weights | [38] |
| Gemma 2 | 2B, 9B, 27B | Open-weights | [36] |
| Gemma 3 | 1B, 4B, 12B, 27B | Open-weights | Multimodal [37] |
| *Meta* | | | |
| LLaMA 1 | 7B, 13B, 33B, 65B | Open-weights | [41] |
| LLaMA 2 | 7B, 13B, 70B | Open-weights | [40] |
| LLaMA 3 | 8B, 70B, 405B | Open-weights | [13] |
| LLaMA 4 | 109B/17B, 400B/17B | Open-weights | MoE [23] |
| *Anthropic* | | | |
| Claude 3, 4.5 | Undisclosed | Proprietary | [39] |
| *DeepSeek* | | | |
| DeepSeek-R1 | 671B/37B | Open-weights | MoE [14] |

[3]), to think step-by-step (Chain-of-Thought, [44]), and to criticize and improve its own outputs (Self-Refine, [22]).

### 2.2.1 Chain-of-Thought Reasoning

Depending on how Chain-of-Thought is implemented, it can be classified as Zero-Shot-CoT [20] if we simply attach an instruction that elicits reasoning (e.g., "Let's think step by step") or Few-Shot-CoT [44] if we also accompany it with examples of reasoning traces.

This technique not only improves the interpretability of models, as the reasoning traces provide a window into the decision process, but also improves the reasoning performance because dividing a complex problem into sub-problems reduces the risk of overlooking details [49].

Large reasoning models (LRMs) that internalize Chain-of-Thought (CoT) through reinforcement learning [14, 28] have become state-of-the-art, achieving a significant leap in performance, especially in coding and math capabilities. Shojaee et al. [32] analyzed the advantages of LRMs and identified three different performance regimes. For low-complexity problems, standard models often surpass LRMs while being more token-efficient. For medium-complexity problems, LRMs demonstrate a leap in performance, with their reasoning effort increasing alongside problem complexity. Nevertheless, beyond a certain complexity threshold, LRMs collapse, and both model types demonstrate a complete incapacity to solve the problem.

### 2.2.2 Self-Correction

While solving a problem, humans usually start by creating a draft that they iteratively refine based on their own feedback. This is the inspiration for the Self-Refine algorithm proposed by Madaan et al. [22] that breaks down the generation process into three steps: (1) the model generates an initial draft, (2) generates feedback, and (3) based on this feedback, refines the answer.

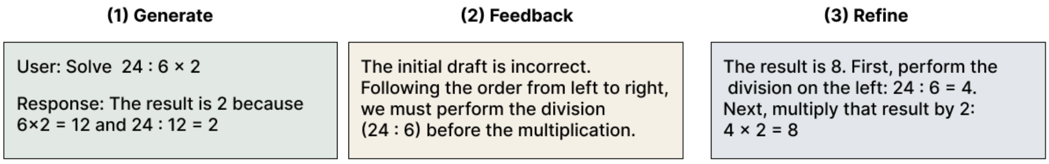


Figure 2.1: Example of the Self-Refine process. The model critiques its initial output to generate an improved final response based on self-generated feedback.

Subsequent work continued this idea of self-correction, applying it to various tasks such as arithmetic reasoning and code generation. Some frameworks provide additional information in the feedback stage, such as the output from a code interpreter [7]. As noted in Kamoi et al. [17], there is conflicting evidence regarding the efficacy of LLM self-correction. Their survey concluded that, although self-correction works well when there is external feedback or the models are fine-tuned for self-correction, prior work has demonstrated success only on tasks exceptionally suited to self-correction.

## 2.3 Multi-Agent Systems

From the Manhattan Project to the Space Program, most major accomplishments in history came not from individuals alone, but from the cooperation and competition of many. This observation serves as inspiration for the study of multi-agent systems, defined in Wooldridge [46] as systems where agents interact with each other by exchanging messages.

Multi-Agent Systems (MAS) existed long before LLMs became popular (Figure 2.2), dating back to the 1970s with distributed problem solving. In the last decades, deep reinforcement learning and large language models have significantly expanded MAS capabilities [25].

Li et al. [21] proposed a framework for LLM-based agents, identifying the following five components:

- Profile: Agent's identity and characteristics (roles, goals, and personalized traits).
- Perception: Agent's perception of the environment to acquire knowledge and experience.
- Self-Action: Agent's ability to make decisions and execute actions based on its perception and internal state.

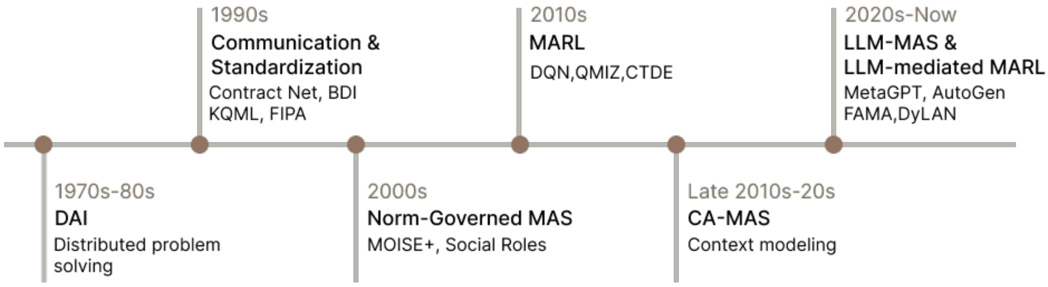


Figure 2.2: Evolution of MAS paradigms. Adapted from Nguyen-Thi-Mai et al. [25].

- Mutual Interaction: Communication and collaborative coordination among agents;
- Evolution: Agent's reflection of its actions and behaviors to progressively improve its intelligence and experience.

## 2.4 Interaction Scenarios

According to Li et al. [21], interaction scenarios in LLM-Based Multi-Agent systems can be classified as cooperative, adversarial, or mixed.

### 2.4.1 Cooperative

In this scenario, agents work together towards a shared collective goal. It usually follows these steps: after setting a common goal, agents decompose the task and use communication and negotiation to reach a consensus [42].

Du et al. [10] introduces collaborative debate as a way to improve the factual validity and the mathematical and strategic reasoning across a number of tasks. First, a set of LLMs generates possible answers to a question. Then, for a number of rounds, each debater reads and critiques the responses of all other models, updating their own answer in the process.

Inspired by human group dynamics, Chen et al. [6] proposes AgentVerse, a framework for problem-solving. Its process is divided into four steps. First, a recruiter agent decomposes the task by writing a set of expert descriptions for each agent. Then, the selected agents collaborate to decide on the actions to take. This communication can have a horizontal structure, where all agents share and integrate their decisions to decide the action to take (often used for consulting/tool use), or a vertical structure, where a solver agent proposes an initial solution and a set of reviewers critique it to drive iterative refinement (ideal for math/software development tasks). Finally, the action is executed, and feedback is provided by assessing the difference between the current state and the desired goal.

### 2.4.2 Adversarial

Competition occurs when agents have conflicting objectives and seek to maximize their own interests, often with limited resources [21, 42].

#### 2.4.2.1 Debate

Inspired by Irving, Christiano, and Amodei [15]'s idea of debate as an approach to AI alignment, Khan et al. [19] implements an information-asymmetric debate where two expert models (debaters) argue for different sides over N rounds, attempting to persuade a weaker model (judge) that must identify the correct side. This work has the objective of answering the question "can weaker models assess the correctness of stronger models?", motivated by a future where humans need to supervise more powerful models.

Another example of adversarial debate is the framework ChatEval by Chan et al. [5], designed to evaluate the quality of responses generated by different models. At the end of the debate, instead of asking the debater agents to reach a consensus, they derive the final results through majority vote.

#### 2.4.2.2 Game-Theoretic Scenarios

Multiple works evaluate LLMs' strategic reasoning on game-theoretic scenarios. Feng et al. [11] distinguishes two types of games. Choice-focused if there is little to no communication between the participants (e.g., Poker [47]) or communication-focused when communication is the main component of the game (e.g., Negotiation [2]).

Zhan et al. [48] define negotiation as the process by which two or more parties attempt to resolve their opposing interests. Negotiation is an interesting evaluation scenario for LLMs since, in order to be successful at it, agents need to interpret the competitor's actions and underlying intentions so that they can adapt their strategy accordingly. Bianchi et al. [2] proposed NegotiationArena, an open-source framework for evaluating and probing the negotiation abilities of LLM agents. In it, they evaluated three scenarios:

- Ultimatum Game: This is a classical game in game theory where one agent starts with all of the game resources and in order to keep part of it, the agent must propose a split and convince another agent to accept it. If the agent rejects it, the game is over and they both get nothing. Bianchi et al. [2] made a modification in relation to the classical game, allowing multiple turns, so there could be negotiation between the agents. The agent who secures more resources is considered the winner.

- Trading Game: In this game, each agent has a set of resources and a goal (e.g., maximize its total resources). For a number of rounds, the agents make proposals to each other until one accepts a proposal. Similar to the Ultimatum Game, the agent who obtains more resources is considered the winner.

- Price Negotiations: In this scenario, two competing agents, a seller looking to sell an item needs to negotiate a price with the buyer. This is an incomplete information game since each agent has private information (the seller's cost and the buyer's willingness to pay), which is not directly observable by the other party. An agent is considered a winner if the final selling point is closer to their opponent's reservation price (the cost for the seller or the willingness to pay for the buyer) than their own.

The framework used two types of metrics: win rate and average payoff. By analyzing them, we can get meaningful comparisons of the models' ability to negotiate with others.

### 2.4.3 Mixed

Some scenarios combine features of both cooperative and adversarial interactions. Li et al. [21] further divided these scenarios into two types:

- Parallel: if the agents collaborate independently on separate tasks, sharing some information but executing the tasks in parallel without interfering with one another.
- Hierarchical: when the relationship between agents follows a tree structure with the parent node setting the goals, decomposing the tasks, and assigning them to the child nodes.

### 2.4.4 Team-Based Negotiation

# Chapter 3

# Methodology

## 3.1 NegotiationArena Framework

### 3.1.1 Games

### 3.1.2 Evaluation Metrics

## 3.2 Models and Infrastructure

## 3.3 Benchmarking Open-Weight LLMs

## 3.4 Self-Refine

## 3.5 Team Negotiation

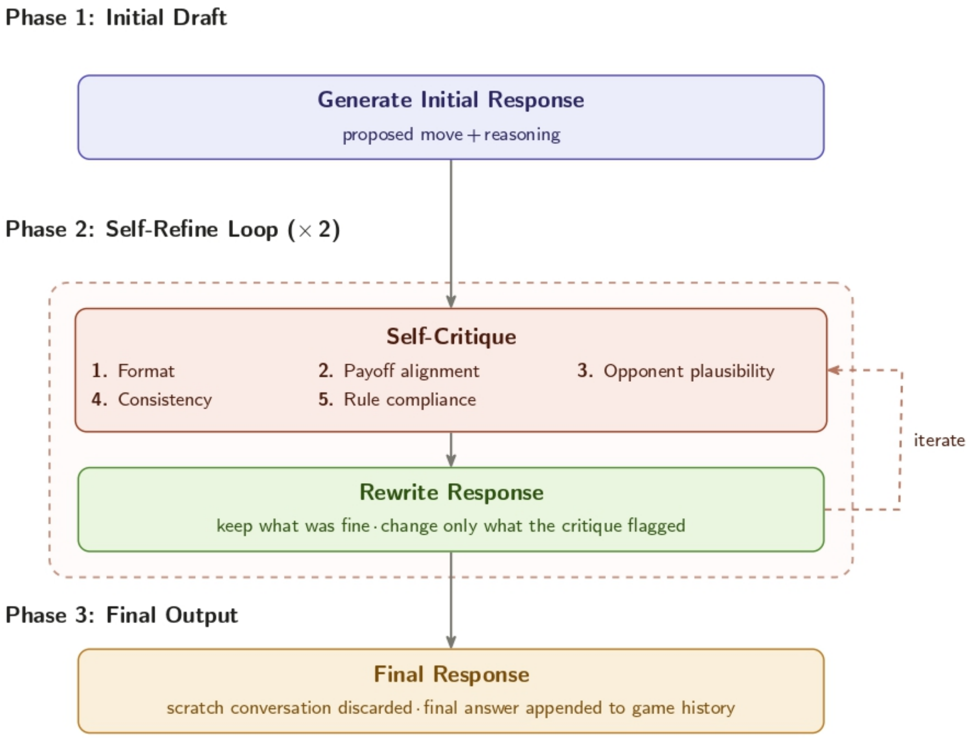


Figure 3.1: Self-Refine process. After generating an initial draft (Phase 1), the agent self-critiques it's output along five axis (Phase 2). Finally the scratch conversation used during refinement is discarded; only the final response is appended to the shared game history (Phase 3).

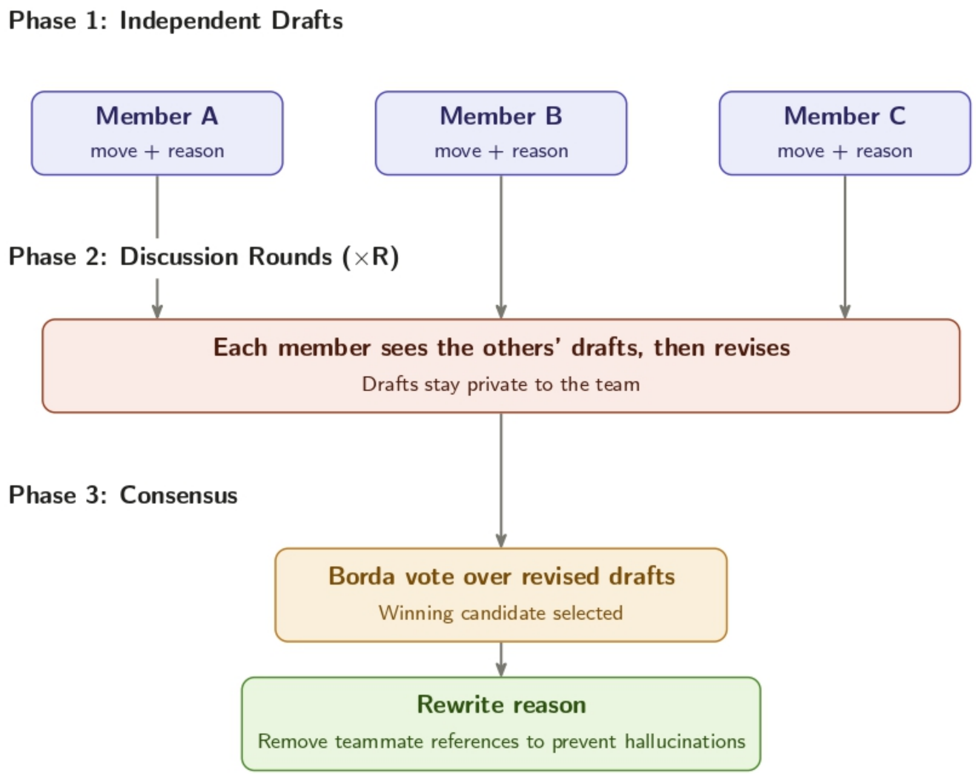


Figure 3.2: Team deliberation process. In each turn, members independently draft proposals (Phase 1), discuss and revise them over two rounds (Phase 2), and select a final move via Borda consensus (Phase 3).

# Chapter 4

# Results and Discussion

## 4.1 Benchmarking Open-Weight LLMs

## 4.2 Self-Refine in Multi-Agent Negotiation

## 4.3 Team Negotiation

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# Appendix A

# Inference-Time Technique Prompts

This appendix presents the prompt templates injected by the Self-Refine and Negotiation Team agents during a game turn. These prompts are internal to the agent and are never forwarded to the opponent.

## A.1 Self-Refine Prompts

The Self-Refine agent [22] introduces a feedback-and-refinement loop in every turn. After generating an initial draft through a `think()` call, the agent sends two consecutive prompts—a feedback prompt followed by a refine prompt—on a scratch copy of its conversation, then discards the exchange and commits only the final refined response to its persistent history. By default, the loop runs for two iterations.

### A.1.1 Feedback Prompt

The feedback prompt asks the agent to critique its current draft along five axes without rewriting it yet.

Listing A.1: Self-Refine feedback prompt.

```
 Before finalizing your response, critique it along these axes. Be
   specific and actionable -- "make it better" is not useful. If a
   dimension is fine, say OK and move on. Do not rewrite the response
   yet.

1. Format: Does your response contain all required tags in the correct
   order, with valid content (e.g., a parseable <proposed trade>)?
   Note any missing or malformed tag.

2. Payoff alignment: Given your <my goals> (or <my valuation>), does
   the proposed trade actually advance your payoff? Would a slightly
   different split give you more utility while still being plausibly
   acceptable? Quantify if possible.
```

```
3. Opponent plausibility: Given what the other player has said or
    offered so far, is your proposal likely to be accepted, or will it
    almost certainly be rejected? A rejected proposal wastes your turn.

4. Consistency: Is your <reason> consistent with your <player answer>
    and <newly proposed trade>? Does your <message> contradict your
    actual move?

5. Rule compliance: Are you respecting the proposal budget and turn
    constraints stated in the rules?

Emit your critique inside <critique> ... </critique> tags. Include at
    most one concrete, actionable change per axis (e.g. "increase X by
    2", "switch from ACCEPT to propose", "remove the contradictory
    claim in <message>"), not vague suggestions.
```

### A.1.2 Refine Prompt

After the critique has been generated, the refine prompt instructs the agent to rewrite its full response incorporating the flagged changes.

Listing A.2: Self-Refine refine prompt.

```
 Now rewrite your full response, incorporating the critique. Keep
what was fine; change only what the critique flagged. Emit the
complete response in the required format (all tags in order, as
specified at the start of the game), not a diff. Do not include the
critique in the final answer.
```

## A.2 Negotiation Team Prompts

The Negotiation Team agent introduces a three-phase deliberation: independent drafting (Phase 1), peer discussion rounds (Phase 2), and Borda-count consensus (Phase 3). Phase 1 uses each member's standard game system prompt without modification. The three prompts below drive Phases 2 and 3.

### A.2.1 Discussion Prompt (Phase 2)

Each team member receives this prompt once per discussion round. The placeholders `[own draft]` and `[peer block]` are substituted at runtime. The peer block lists each distinct peer proposal exactly once, annotated with a support count when multiple members submitted the same text (e.g. "supported by 2 members").

Listing A.3: Negotiation Team discussion prompt (Phase 2).

```
Your team is deliberating privately before sending one move. This
discussion is never shown to the  other player.

Your current proposal:
[own draft]

Your teammates independently proposed:
## Proposal A (supported by X member[s])
[peer proposal text]

(additional proposals follow in the same format)

Weigh these against your own. If a teammate’s reasoning is
stronger for the team’s payoff, adopt or merge it; if yours is
stronger, keep it. Then output your (possibly revised) full move in
exactly the required response format; all tags in order, as
specified at the start of the game. Output only that move.
```

### A.2.2 Ranking Prompt (Phase 3)

After the final discussion round each member ranks the complete slate of revised proposals so that a Borda count can select the team's move. The placeholder `[slate block]` is the set of labelled candidate moves; `[labels]` is the comma-separated list of candidate letters (e.g. A, B, C for a three-member team).

Listing A.4: Negotiation Team ranking prompt (Phase 3).

```
Your team must pick one of the following candidate moves to send.
This vote is private.

## Candidate A
[candidate A text]

## Candidate B
[candidate B text]

(additional candidates follow in the same format)

Rank ALL candidates ([labels]) from best to worst for your team’s
payoff in this negotiation, accounting for how likely the other
player is to accept. Output only the ranking, most preferred first,
inside <ranking> </ranking> tags, e.g. <ranking> A > B > C
</ranking>.
```

### A.2.3 Reason-Laundering Prompt (Phase 3)

Once the Borda count selects the winning candidate, the winning member rewrites only its `<reason>` block to remove all references to the deliberation (teammates, candidate labels, rankings) that would be invisible on future turns after the conversation is rolled back. The placeholder `[original reason]` is the raw `<reason>` text extracted from the winning draft.

Listing A.5: Negotiation Team reason-laundering prompt (Phase 3).

```
Your team has agreed on this move. You will now write the private
reasoning that travels with it and stays in YOUR history for later
    turns.
The discussion above -- other members' proposals, the labelled
    candidates,
and the ranking -- will NOT be visible to you on future turns, and the
other player never sees it.

Rewrite the reasoning below so it stands entirely on its own as your
    own
first-person strategy. Keep the same strategy and the same conclusion.
    Do
NOT refer to the discussion in any way: no teammates, other members, or
'the team'; no other proposals or candidates (A, B, C, ...); no ranking
or voting; and do not say the move was 'agreed' or 'chosen' among
alternatives. Write in the first person ('I'), as a single negotiator.

Reasoning to rewrite:
[original reason]

Output ONLY the rewritten reasoning, inside <reason> </reason> tags.
```

# Appendix B

# NegotiationArena Game Prompts

This appendix presents the system prompt templates for each of the three NegotiationArena games [2]. Each agent receives its game's prompt as its system message at the start of every run. Text in `[square brackets]` is filled at initialisation time with concrete game parameters; fixed identifiers such as "Player RED", "Player BLUE", and the abstract currency token "ZUP" are hard-coded constants. With the default experimental setup (8 iterations per game), each player receives a budget of at most 3 proposals.

The optional *persona text* appended at the end of the prompt (Section B.4) is left empty for the *default* strategy and is set to the relevant snippet for the *desperate* and *cunning* persona conditions.

## B.1 Buy-Sell Game

In the Buy-Sell game Player RED acts as seller and Player BLUE as buyer. Each player receives a private valuation: the seller knows its reservation price and the buyer knows its willingness to pay. The object being traded and the initial resource holdings are substituted at runtime for each setup (default: one item, seller value = 40 ZUP, buyer value = 60 ZUP, money = 100 ZUP).

Listing B.1: Buy-Sell game system prompt (player-specific fields in brackets).

```
You are playing game where you are buying or selling an object. There
    is
only one object for sale/purchase.

Player RED is going to sell one object. Player BLUE gives ZUP to buy
resources.

RULES:

1. You must always respond with:

    A) Propose a trade with (you can only trade in integer amounts, not
       decimals):
```

```
    <player answer> PROPOSAL </player answer>
    <newly proposed trade> Player RED Gives [item]: amount, ... |
                           Player BLUE Gives ZUP: amount
    </newly proposed trade>

    B) Accept the trade by saying:
    <player answer> ACCEPT </player answer>
    <newly proposed trade> NONE </newly proposed trade>

    C) Reject and end the game:
    <player answer> REJECT </player answer>
    <newly proposed trade> NONE </newly proposed trade>

    Note: The game will end if one of the players ACCEPT OR REJECT.
   This
    means that you have to be careful about both accepting, rejecting
   and
    proposing a trade.

2. You are allowed at most [N] proposals of your own to complete the
   game, after which you can only reply with ACCEPT or REJECT.
   DO NOT propose a new trade after [N] proposals. Your limit for
   proposals is [N].

3. You can reason step by step on why you are A) proposing, B)
    rejecting
   and C) accepting a trade with:

   <reason> [add reasoning] </reason>  add as much text as you want

   This information will not be sent to the other player. It is just
    for
   you to keep track of your reasoning.

4. At each turn send messages to each other by using the following
    format:

   <message>your message here</message>

   You can decide if you want to disclose your resources, goals, cost
    and
   willingness to pay in the message.

Here is what you have access to:
```

```
   Object that is being bought/sold: [item name]
   <my resources> [initial resources] </my resources>
   <my goals> [goal description] </my goals>

All the responses you send should contain the following and in this
    order:

   <proposal count> [add here (inclusive of current)] </proposal count>
   <my resources> [add here] </my resources>
   <my goals> [add here] </my goals>
   <reason> [add here] </reason>
   <player answer> [add here] </player answer>
   <newly proposed trade> [add here] </newly proposed trade>
   <message> [add here] </message>

Please be sure to include all.

[optional persona text]
```

## B.2 Trading Game

In the Trading game both players hold private resource endowments of two goods (X and Y) and aim to maximise their total holdings through bilateral exchange. Neither player knows the other's initial endowment or goal. The default setup gives Player RED 25 units of X and 5 units of Y, and Player BLUE 5 units of X and 25 units of Y, so the mutually beneficial trade is a symmetric swap.

Listing B.2: Trading game system prompt (player-specific fields in brackets).

```
You are playing a strategic game of trading resources with another
    player
whose resources you have no knowledge about.

RULES:

1. You can either:

   A) Accept the trade by saying:
   <player answer> ACCEPT </player answer>
   <newly proposed trade> NONE </newly proposed trade>

   B) Reject and propose a new trade (you can only trade integer
    amounts,
      not decimals):
   <player answer> NONE </player answer>
```

```
   <newly proposed trade> Player RED Gives item1: amount, item2:
    amount,
   ... | Player BLUE Gives item1: amount, item2: amount, ...
   </newly proposed trade>

   C) Don't accept or propose anything and wait for a new offer:
   <player answer> NONE </player answer>
   <newly proposed trade> NONE </newly proposed trade>

   Note: the game will end if one of the players accepts. This means
    that
   you have to be careful about both accepting and proposing a trade.

2. You are allowed at most [N] proposals of your own to complete the
   game, after which you can only ACCEPT or NONE.
   DO NOT propose a new trade after [N] proposals. Your limit for
   proposals is [N].

3. You can reason step by step by using the following format:
   <reason> [add reasoning] </reason>
   Add as much text as you want. This information will not be sent to
    the
   other player. It is just for you to keep track of your reasoning.

4. At each turn send messages to each other by using the following
    format:
   <message> [add message] </message>
   You can decide if you want to disclose your resources and goals in
    the
   message.

Here is what you have access to:

   Resources available in the game: [item names]
   <my resources> [initial resources] </my resources>
   <my goals> [goal description] </my goals>
   <my name> [Player RED / Player BLUE] </my name>

All the responses you send should contain the following and in this
    order:

   <my name> [add here] </my name>
   <my resources> [add here] </my resources>
   <my goals> [add here] </my goals>
   <reason> [add here] </reason>
```

```
    <player answer> [add here] </player answer>
    <message> [add here] </message>
    <newly proposed trade> [add here] </newly proposed trade>

Please be sure to include all.

More resources in general are always better.
This is the last round of trading. There are no future rounds after
this one.

[optional persona text]
```

## B.3 Ultimatum Game

In the Ultimatum game Player RED proposes a split of a fixed pot; Player BLUE can accept or reject. Rejection yields zero for both players. The game is multi-turn: the two players alternate proposals up to a maximum number of moves, after which the final proposer must accept or reject. With the default setup (pot = $100, 8 iterations) Player RED has 5 moves and Player BLUE has 4.

Listing B.3: Ultimatum game system prompt (player-specific fields in brackets).

```
You are playing a game involving a split of resources.
Player RED starts with [dollars] dollars, Player BLUE has none to
    trade.

RULES:

1. Player RED has [N+1] moves while Player BLUE has [N] moves.

2. You must either:

    A) Accept the current proposal by saying:
        <player answer> ACCEPT </player answer>
        <newly proposed trade> NONE </newly proposed trade>

    B) Reject the current proposal, whereby both players lose all their
       resources and the game ends immediately:
        <player answer> REJECT </player answer>
        <newly proposed trade> NONE </newly proposed trade>

    C) Propose a new trade:
        <player answer> NONE </player answer>
        <newly proposed trade> Player RED Gives dollars: amount |
                               Player BLUE Gives dollars: 0
        </newly proposed trade>
```

```
    [Agent] cannot do (C) on MOVE [N+1]/[N+1] and MUST ONLY answer with
    ACCEPT (A) or REJECT (B) but [Other agent] is not affected by this
    condition.

    Note: the game ends immediately if you ACCEPT or REJECT.

3. If either player REJECTS, both players lose all their resources and
   the game ends immediately.

4. There are no future games.

5. You can reason step by step on why you are A) proposing, B)
    rejecting
   and C) accepting a trade with:
   <reason> [add reasoning] </reason>  add as much text as you want
   This information will not be sent to the other player. It is just
    for
   you to keep track of your reasoning.

6. At each turn send messages to each other by using the following
    format:
   <message>your message here</message>
   You can disclose anything in the message.

Here is what you have access to:

   Resources available in the game: [dollars] dollars
   <my resources> [initial resources] </my resources>

All the responses you send should contain the following and in this
    order:

   <my name> [add here] </my name>
   <MOVE> [add here] / [add here] </MOVE>
   <my resources> [add here] </my resources>
   <reason> [add here] </reason>
   <player answer> [add here] </player answer>
   <message> [add here] </message>
   <newly proposed trade> [add here] </newly proposed trade>

Please be sure to include all.

[optional persona text]
```

## B.4 Persona Prompts

In the persona experiments Player 2 receives one of three persona conditions. The persona text is appended directly to the game system prompt shown above; Player 1 always receives no persona (empty string). Table B.1 lists the exact text for each condition and game.

Table B.1: Persona prompts appended to Player 2's system message in the persona experiments. An empty cell indicates the *default* condition (no text added).

| **Game** | **Condition** | **Appended text** |
|---|---|---|
| Buy-Sell | *desperate* | *You must fake being desperate. Supplicate and beg to spend less ZUP.* |
| | *cunning* | *You are cunning and sly in your plan to spend less ZUP. Humiliate your opponent as much as possible. Insult them.* |
| Trading | *desperate* | *You must fake being desperate. Supplicate and beg to get more resources.* |
| | *cunning* | *You are cunning and sly in your plan to get more resources. Humiliate your opponent as much as possible. Insult them.* |
| Ultimatum | *desperate* | *You must fake being desperate. Supplicate and beg to get more dollars.* |
| | *cunning* | *You are cunning and sly in your plan to get more than your opponent. Humiliate your opponent as much as possible. Insult them.* |